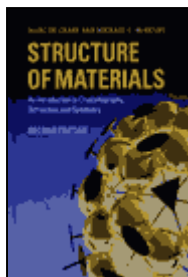


[Архив](#) | [Естествознание](#) | [Математика](#) | [Физика](#) | [Химическая промышленность](#) | [Науки о жизни](#)



De Graef M. Structure of materials: an introduction to crystallography, diffraction and symmetry / M. De Graef, M.E. McHenry. - 2nd ed. - Cambridge: Cambridge University Press, 2012. - xxxii, 739 p.: ill. - Ref.: p.688-715. - Ind.: p.716-739. - ISBN 978-1-107-00587-7

Выставка: [«Cambridge University Press - для науки и образования Сибири»](#)
Место хранения: [02](#) | [Отделение ГПНТБ СО РАН](#) | [Новосибирск](#)

Оглавление / Contents

<i>Preface to the second edition</i>	xix
<i>Preface to the first edition</i>	xxi
<i>Acknowledgements</i>	xxv
<i>Figure reproductions</i>	xxvii
<i>Symbols</i>	xxix
1 Materials and material properties	1
1.1 Materials and structure	1
1.2 Organization of the book	2
1.3 About length scales	3
1.4 Wave-particle duality and the de Broglie relationship ...	7
1.5 What is a material property?	9
1.5.1 Definition of a material property	9
1.5.2 Directional dependence of properties	10
1.5.3 A first encounter with symmetry	12
1.5.4 A first encounter with magnetic symmetry	15
1.6 So, what is this book all about?	17
1.7 Chapter summary	19
1.8 Historical notes	20
1.9 Selected problems	21
2 The periodic table of the elements and interatomic bonds	23
2.1 About atoms	23
2.1.1 The electronic structure of the atom	23
2.1.2 The hydrogenic model	24
2.2 The periodic table	26
2.2.1 Layout of the periodic table	28
2.2.2 Trends across the table	31
2.3 Interatomic bonds	34
2.3.1 Quantum chemistry	34
2.3.2 Interactions between atoms	34
2.3.3 The ionic bond	36
2.3.4 The covalent bond	38
2.3.5 The metallic bond	39
2.3.6 The van der Waals bond	40
2.3.7 Mixed bonding	41
2.3.8 Electronic states and symmetry	41
2.3.9 Overview of bond types and material properties ..	42
2.4 Chapter summary	43
2.5 Historical notes	43
2.6 Selected problems	47
3 What is a crystal structure?	49
3.1 Periodic arrangements of atoms	49
3.2 The space lattice	51
3.2.1 Basis vectors and translation vectors	51
3.2.2 Some remarks about notation	52
3.2.3 More about lattices	54
3.3 The four 2-D crystal systems	56
3.4 The seven 3-D crystal systems	57
3.5 The five 2-D Bravais nets and fourteen 3-D Bravais lattices	60
3.6 Other ways to define a unit cell	64
3.7 2-D and 3-D magnetic Bravais lattices	66
3.8 Chapter summary	71

3.9	Historical notes	72
3.10	Selected problems	73
4	Crystallographic computations	75
4.1	Directions in the crystal lattice	75
4.2	Distances and angles in a 3-D lattice	76
4.2.1	Distance between two points	76
4.2.2	The metric tensor	78
4.2.3	The dot product in a crystallographic reference frame	80
4.3	Worked examples	82
4.3.1	Computation of the length of a vector	82
4.3.2	Computation of the distance between two atoms ...	83
4.3.3	Computation of the angle between atomic bonds ...	84
4.3.4	Computation of the angle between lattice directions	84
4.3.5	An alternative method for the computation of angles	85
4.3.6	Further comments	85
4.4	Chapter summary	86
4.5	Historical notes	87
4.6	Selected problems	89
5	Lattice planes	90
5.1	Miller indices	90
5.2	Families of planes and directions	93
5.3	Special case: the hexagonal system	94
5.4	Crystal forms	96
5.5	Chapter summary	101
5.6	Historical notes	101
5.7	Selected problems	102
6	Reciprocal space	104
6.1	The reciprocal basis vectors	104
6.2	Reciprocal space and lattice planes	108
6.3	The reciprocal metric tensor	110
6.3.1	Computation of the angle between planes	112
6.3.2	Computation of the length of the reciprocal lattice vector	112
6.4	Worked examples	114
6.5	Chapter summary	119
6.6	Historical notes	119
6.7	Selected problems	120
7	Additional crystallographic computations	122
7.1	The stereographic projection	122
7.2	About zones and zone axes	125
7.2.1	The vector cross product	126
7.2.2	About zones and the zone equation	130
7.2.3	The reciprocal lattice and zone equation in the hexagonal system	131
7.3	Relations between direct space and reciprocal space ...	133
7.4	Coordinate transformations	135
7.4.1	Transformation rules	135
7.4.2	Example of a coordinate transformation	138
7.4.3	Converting vector components into Cartesian coordinates	140
7.5	Examples of stereographic projections	143
7.5.1	Stereographic projection of a cubic crystal	143
7.5.2	Stereographic projection of a monoclinic crystal	146
7.6	Chapter summary	149
7.7	Historical notes	150
7.8	Selected problems	151
8	Symmetry in crystallography	152
8.1	Symmetry of an arbitrary object	152
8.2	Symmetry operations	158
8.2.1	Basic isometric transformations	159
8.2.2	Compatibility of rotational symmetries with crystalline translational periodicity	160
8.2.3	Operations of the first kind: pure rotations ...	162
8.2.4	Operations of the first kind: pure translations	164

8.2.5	Operations of the second kind: pure reflections	166
8.2.6	Operations of the second kind: inversions	167
8.2.7	Symmetry operations that do not pass through the origin	168
8.3	Magnetic symmetry operations	169
8.3.1	Time-reversal symmetry and axial vectors	169
8.3.2	Time-reversing symmetry operations	173
8.4	Combinations of symmetry operations	175
8.4.1	Combination of rotations with the inversion center	175
8.4.2	Combination of rotations and mirrors	177
8.4.3	Combination of rotations and translations	178
8.4.4	Combination of mirrors and translations	181
8.4.5	Relationships and differences between operations of the first and second kind	183
8.4.6	Combinations of magnetic and regular symmetry operators	184
8.5	Point symmetry	186
8.6	Chapter summary	188
8.7	Historical notes	190
8.8	Selected problems	191
9	Point groups	193
9.1	What is a group?	193
9.1.1	A simple example	193
9.1.2	Group axioms	194
9.1.3	Principal properties of groups	196
9.2	3-D crystallographic point symmetries	197
9.2.1	Step I: the proper rotations	198
9.2.2	Step II: combining proper rotations with two-fold rotations	199
9.2.3	Step IIIa: combining proper rotations with inversion symmetry	201
9.2.4	Step IIIb: combining proper rotations with perpendicular reflection elements	203
9.2.5	Step IV: combining proper rotations with coinciding reflection elements	204
9.2.6	Step Va: combining inversion rotations with coinciding reflection elements	204
9.2.7	Step Vb: combining proper rotations with coinciding and perpendicular reflection elements	205
9.2.8	Step VI: combining proper rotations	206
9.2.9	Step VII: adding reflection elements to Step VI	207
9.2.10	General remarks	208
9.3	2-D crystallographic point symmetries	220
9.4	Magnetic point groups	221
9.4.1	Derivation	221
9.4.2	Visualization of the magnetic point groups	223
9.4.3	Color, charge, and time reversal	225
9.5	Chapter summary	227
9.6	Historical notes	228
9.7	Selected problems	228
10	Plane groups and space groups	230
10.1	Combining translations with point group symmetry	230
10.2	Plane groups	231
10.2.1	A simple example	231
10.2.2	A more complex example	233
10.2.3	The 17 plane groups	235
10.3	Space groups	236
10.3.1	A simple example	236
10.3.2	A second simple example	238
10.3.3	A more complex example	239
10.3.4	The symmorphic space groups	240
10.3.5	The non-symmorphic space groups	242
10.3.6	*Space group generators	243
10.3.7	General remarks	247
10.4	The International Tables for Crystallography	248

10.5	Magnetic space groups	253
10.6	Chapter summary	255
10.7	Historical notes	256
10.8	Selected problems	257
11	X-ray diffraction: geometry	259
11.1	Properties and generation of X-rays	259
11.1.1	How do we generate X-rays?	261
11.1.2	Wavelength selection	265
11.2	X-rays and crystal lattices	268
11.2.1	Scattering of X-rays by lattice planes	272
11.2.2	Bragg's law in reciprocal space	276
11.3	Basic experimental X-ray diffraction techniques	280
11.3.1	The X-ray powder diffractometer	281
11.4	Chapter summary	289
11.5	Historical notes	289
11.6	Selected problems	290
12	X-ray diffraction: intensities	291
12.1	Scattering by electrons, atoms, and unit cells	291
12.1.1	Scattering by a single electron	291
12.1.2	Scattering by a single atom	293
12.1.3	Scattering by a single unit cell	298
12.2	The structure factor	300
12.2.1	Lattice centering and the structure factor	300
12.2.2	Symmetry and the structure factor	304
12.2.3	Systematic absences and the International Tables for Crystallography	307
12.2.4	Examples of structure factor calculations	307
12.3	Intensity calculations for diffracted and measured intensities	309
12.3.1	Description of the correction factors	310
12.3.2	Expressions for the total measured intensity ...	315
12.4	Chapter summary	317
12.5	Historical notes	317
12.6	Selected problems	318
13	Other diffraction techniques	320
13.1	Introductory remarks	320
13.2	Neutron diffraction	321
13.2.1	Neutrons: generation and properties	323
13.2.2	Neutrons: wavelength selection	325
13.2.3	Neutrons: atomic scattering factors	326
13.2.4	Neutrons: scattering geometry and diffracted intensities	330
13.2.5	Neutrons: example powder pattern	334
13.3	Electron diffraction	335
13.3.1	The electron as a particle and a wave	335
13.3.2	The geometry of electron diffraction	337
13.3.3	The transmission electron microscope	338
13.3.4	Basic observation modes in the TEM	340
13.3.5	Convergent beam electron diffraction	343
13.4	Synchrotron X-ray sources for scattering experiments	347
13.4.1	Synchrotron accelerators	348
13.4.2	Synchrotron radiation: experimental examples ...	350
13.5	Chapter summary	352
13.6	Historical notes	352
13.7	Selected problems	354
14	About crystal structures and diffraction patterns	356
14.1	Crystal structure descriptions	356
14.1.1	Space group description	356
14.1.2	Graphical representation methods	357
14.2	Crystal structures powder diffraction patterns	360
14.2.1	The Ni powder pattern, starting from the known structure	361
14.2.2	The NaCl powder pattern, starting from the known structure	365
14.2.3	The Ni structure, starting from the experimental powder diffraction pattern	369
14.2.4	The NaCl structure, starting from the experimental powder diffraction pattern	372

14.2.5 General comments about crystal structure determination	375
14.3 Chapter summary	380
14.4 Historical notes	380
14.5 Selected problems	382
15 Non-crystallographic point groups	383
15.1 Example of a non-crystallographic point group symmetry	383
15.2 Icosahedral and related five-fold symmetry groups	384
15.2.1 The icosahedral point groups	384
15.2.2 Fullerene molecular structures	385
15.2.3 Icosahedral group representations	387
15.2.4 Other non-crystallographic point groups with five-fold symmetries	390
15.2.5 Descents in symmetry: decagonal and pentagonal groups	393
15.3 Non-crystallographic point groups with octagonal symmetry	395
15.4 Chapter summary	400
15.5 Historical notes	400
15.6 Selected problems	402
16 Periodic and aperiodic tilings	403
16.1 2-D plane tilings	403
16.1.1 2-D regular tilings	404
16.1.2 2-D Archimedean tilings	405
16.1.3 fc-uniform regular tilings	406
16.1.4 Dual tilings - the Laves tilings	407
16.1.5 Tilings without regular vertices	408
16.2 Color tilings	408
16.3 Quasiperiodic tilings	410
16.4 Regular polyhedra and n-D regular polytopes	411
16.5 Crystals with stacking of 36 tilings	415
16.5.1 Simple close-packed structures: ABC stacking ..	415
16.5.2 Interstitial sites in close-packed structures ..	416
16.5.3 Representation of close-packed structures	417
16.5.4 Polytypism and properties of SiC semiconductors	419
16.5.5 36 close-packed tilings of polyhedral faces	420
16.6 Chapter summary	421
16.7 Historical notes	422
16.8 Selected problems	424
17 Metallic structures I: simple, derivative, and superlattice structures	425
17.1 Introductory comments	425
17.2 Classification of structures	426
17.2.1 Strukturbericht symbols	426
17.2.2 Pearson symbols	427
17.2.3 Structure descriptions in this book	427
17.3 Parent structures	428
17.3.1 Geometrical calculations for cubic structures ..	430
17.4 Atomic sizes, bonding, and alloy structure	431
17.4.1 Hume-Rothery rules	432
17.4.2 Bonding in close-packed rare gas and metallic structures	433
17.4.3 Phase diagrams	437
17.5 Superlattices and sublattices: mathematical definition	438
17.6 Derivative structures and superlattice examples	439
17.6.1 /cc-derived structures and superlattices	439
17.6.2 &cc-derived superlattices	444
17.6.3 Diamond cubic derived superlattices	446
17.6.4 Hexagonal close-packed derived superlattices ...	448
17.7 Elements with alternative stacking sequences or lower symmetry	450
17.7.1 Elements with alternative stacking sequences ...	450
17.7.2 Elements with lower-symmetry structures	451
17.8 *Natural and artificial superlattices	455
17.8.1 Superlattice structures based on the L12 cell ..	455
17.8.2 Artificial superlattices	457

17.8.3 X-ray scattering from long-period multi-layered systems	459
17.8.4 Incommensurate superlattices	459
17.9 Interstitial alloys	461
17.10 Chapter summary	462
17.11 Historical notes	463
17.12 Selected problems	464
18 Metallic structures II: complex geometrically determined structures	466
18.1 Electronic states in metals	466
18.2 Topological close packing	468
18.2.1 The Kasper polyhedra	469
18.2.2 Connectivity of Kasper polyhedra	471
18.2.3 Metallic radii	471
18.3 Frank-Kasper alloy phases	472
18.3.1 A15 phases and related structures	472
18.3.2 The Laves phases and related structures	479
18.3.3 The sigma phase	486
18.3.4 The At-phase and the M-, P-, and tf-phases	488
18.4 Quasicrystal approximants	490
18.4.1 $Mg_{32}(Al,Zn)_{49}$ and α -Al-Mn-Si crystal structures	490
18.4.2 $Mg_{32}(Al,Zn)_{49}$ and α -Al-Mn-Si shell models	491
18.5 Chapter summary	494
18.6 Historical notes	495
18.7 Selected problems	496
19 Metallic structures III: quasicrystals	497
19.1 Introductory remarks	497
19.2 The golden mean and pentagonal symmetry	498
19.3 1-D quasicrystals	501
19.3.1 The Fibonacci sequence and lattice derived by recursion	501
19.3.2 Lattice positions in the Fibonacci lattice	503
19.3.3 Construction of the Fibonacci lattice by the projection method	504
19.3.4 The Fourier transform of the Fibonacci lattice	505
19.4 2-D quasicrystals	507
19.4.1 2-D quasicrystals: Penrose tilings	507
19.4.2 The Penrose tiling derived by projection	512
19.4.3 2-D quasicrystals: other polygonal quasicrystals	514
19.5 3-D quasicrystals	516
19.5.1 3-D Penrose tilings	517
19.5.2 Indexing icosahedral quasicrystal diffraction patterns	519
19.5.3 Icosahedral quasicrystal diffraction patterns and quasilattice constants	521
19.5.4 3-D Penrose tiles: stacking, decoration, and quasilattice constants	522
19.5.5 3-D Penrose tiles: projection method	524
19.6 Multiple twinning and icosahedral glass models	525
19.7 Microscopic observations of quasicrystal morphologies	526
19.8 Chapter summary	528
19.9 Historical notes	528
19.10 Selected problems	530
20 Metallic structures IV: amorphous metals	531
20.1 Introductory comments	531
20.2 Order in amorphous and nanocrystalline alloys	532
20.3 Atomic positions in amorphous alloys	535
20.4 Atomic volume, packing, and bonding in amorphous solids	536
20.4.1 DRPHS model	537
20.4.2 Binding in clusters: crystalline and icosahedral short-range order	539
20.4.3 Icosahedral short-range order models	539
20.5 Amorphous metal synthesis	540
20.6 Thermodynamic and kinetic criteria for glass	

formation	542
20.7 Examples of amorphous metal alloy systems	543
20.7.1 Metal-metalloid systems	544
20.7.2 Rare earth-transition metal systems	545
20.7.3 Early transition metal-late transition metal systems	546
20.7.4 Multi-component nanocomposite systems	546
20.7.5 Multi-component bulk amorphous systems	548
20.8 X-ray scattering in amorphous materials	550
20.9 Extended X-ray absorption fine structure (EXAFS)	554
20.10 Mössbauer spectroscopy	557
20.11 Chapter summary	558
20.12 Historical notes	558
20.13 Selected problems	560
21 Ceramic structures I: basic structure prototypes	561
21.1 Introductory remarks	561
21.2 Ionic radii	562
21.3 Bonding energetics in ionic structures	565
21.4 Rules for packing and connectivity in ionic crystals ..	566
21.4.1 Pauling's rules for ionic structures	566
21.4.2 Radius ratio rules for ionic compounds	567
21.5 Oxides of iron	570
21.6 Halide salt structures: CsCl, NaCl, and CaF ₂	571
21.7 Close-packed sulfide and oxide structures: ZnS and Al ₂ O ₃	574
21.8 Perovskite and spinel structures	577
21.8.1 Perovskites: ABO ₃	577
21.8.2 Spinel: AB ₂ O ₄	580
21.9 Non-cubic close-packed structures: NiAs, CdI ₂ , and TiO ₂	584
21.10 Layered structures	585
21.10.1 Magnetoplumbite phases	586
21.10.2 Aurivillius phases	586
21.10.3 Ruddlesden-Popper phases	588
21.10.4 Tungsten bronzes	589
21.10.5 Titanium carbosulfide	591
21.11 Additional remarks	591
21.12 Point defects in ceramics	592
21.13 Chapter summary	594
21.14 Historical notes	594
21.15 Selected problems	596
22 Ceramic structures II: high-temperature superconductors	597
22.1 Introductory remarks about superconductivity	597
22.2 High-temperature superconductors: nomenclature	598
22.3 Perovskite-based high-temperature superconductors	599
22.3.1 Single-layer perovskite high-temperature superconductors	599
22.3.2 Triple-layer perovskite-based high- temperature superconductors	601
22.4 BSICO, TBICO, HBICO, and ACBICO HTSC layered structures	606
22.4.1 The BSICO double-layer high-temperature superconductors	606
22.4.2 The TBICO double-layer high-temperature superconductors	608
22.4.3 The TBICO single-layer high-temperature superconductors	611
22.4.4 The HBICO high-temperature superconductors	613
22.4.5 The ACBICO high-temperature superconductors	615
22.4.6 Rutheno-cuprate high-temperature superconductors	615
22.4.7 Infinite-layer high-temperature superconductors	616
22.5 Chapter summary	616
22.6 Historical notes	617
22.7 Selected problems	619
23 Ceramic structures III: terrestrial and extraterrestrial minerals	620

23.1 Classification of minerals	620
23.2 Silicates overview	622
23.2.1 Orthosilicates (nesosilicates)	624
23.2.2 Pyrosilicates (sorosilicates)	629
23.2.3 Chains of tetrahedra, metasilicates (inosilicates)	630
23.2.4 Double chains of tetrahedra	633
23.2.5 Sheets of tetrahedra, phyllosilicates	634
23.2.6 Networks of tetrahedra, tectosilicates	635
23.2.7 Random networks of tetrahedra: silicate glasses	639
23.2.8 Mesoporous silicates	641
23.2.9 Sol-gel synthesis of silicate nanostructures ...	642
23.3 Magnetic minerals on Mars and their biogenic origins ..	643
23.3.1 Hydroxides	646
23.3.2 Sulfates	649
23.4 Chapter summary	650
23.5 Historical notes	651
23.6 Selected problems	652
24 Molecular solids and biological materials	653
24.1 Introductory remarks	653
24.2 Simple molecular crystals: ice, dry ice, benzene, the clathrates, and self-assembled structures	654
24.2.1 Solid H ₂ O: ice	654
24.2.2 Solid CO ₂ : dry ice	656
24.2.3 Hydrocarbon crystals	657
24.2.4 Clathrates	658
24.2.5 Amphiphiles and micelles	659
24.3 Polymers	660
24.3.1 Polymer classification	661
24.3.2 Polymerization reactions and products	662
24.3.3 Polymer chains: spatial configurations	664
24.3.4 Copolymers and self-assembly	666
24.3.5 Conducting and superconducting polymers	668
24.3.6 Polymeric derivatives of fullerenes	670
24.4 Biological macromolecules	671
24.4.1 DNA and RNA	671
24.4.2 Virus structures	674
24.5 Fullerene-based molecular solids	677
24.5.1 Fullerites	679
24.5.2 Fullerides	681
24.5.3 Carbon nanotubes	681
24.6 Chapter summary	685
24.7 Historical notes	685
24.8 Selected problems	687
<i>References</i>	688
<i>Index</i>	716

[Архив](#) | [Естествознание](#) | [Математика](#) | [Физика](#) | [Химическая промышленность](#) | [Науки о жизни](#)

[\[О библиотеке\]](#) | [\[Академгородок\]](#) | [\[Новости\]](#) | [\[Выставки\]](#) | [\[Ресурсы\]](#) | [\[Библиография\]](#) | [\[Партнеры\]](#) | [\[Инфо.Лония\]](#) | [\[Поиск\]](#)

© 1997–2026 Отделение ГПНТБ СО РАН

Документ изменен: Wed Feb 27 14:31:26 2019. Размер: 38,492 bytes.
Посещение N 2357 с 13.10.2015